

## Results

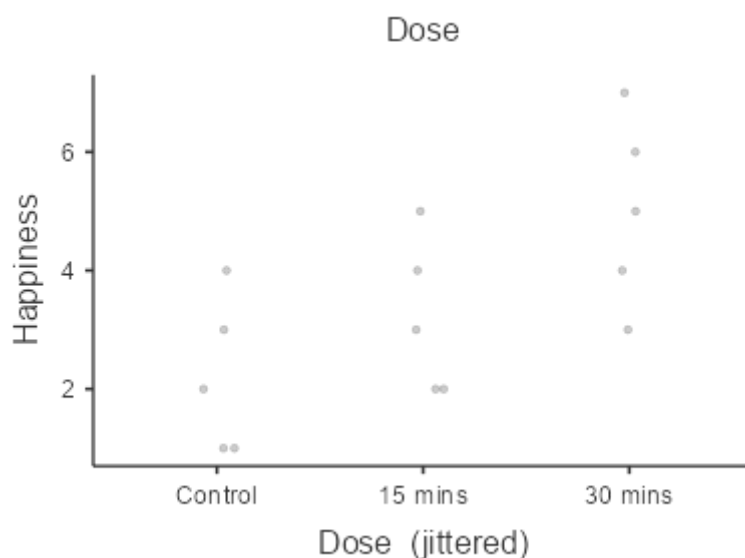
### Relationships, Prediction, and Group Comparisons

You have entered a numeric variable for Variable 1 / Dependent Variable and a nominal variable for Variable 2 / Independent Variables. Hence, a [one way ANOVA](#), which is a test for the difference between several population means, seems to be a good option for you! In order to run this analysis in jamovi, go to: ANOVA > ANOVA

- Drop your dependent (numeric) variable in the box below Dependent Variable and your independent (grouping) variable in the box below Fixed Factors

If the normality or homoscedasticity assumption is violated, you could use the non-parametric [Kruskal-Wallis test](#). Click on the links to learn more about these tests!

### Scatter Plots of Bivariate Relationships - Dependent/Independent Variables



## ANOVA

ANOVA - Happiness

|                  | Sum of Squares | df | Mean Square | F    | p     | $\omega^2$ |
|------------------|----------------|----|-------------|------|-------|------------|
| <b>Dose</b>      | 20.1           | 2  | 10.07       | 5.12 | 0.025 | 0.354      |
| <b>Residuals</b> | 23.6           | 12 | 1.97        |      |       |            |

[3]

### Assumption Checks

Homogeneity of Variances Tests

|                   | Statistic | df | df2 | p     |
|-------------------|-----------|----|-----|-------|
| <b>Levene's</b>   | 0.0917    | 2  | 12  | 0.913 |
| <b>Bartlett's</b> | 0.185     | 2  |     | 0.912 |

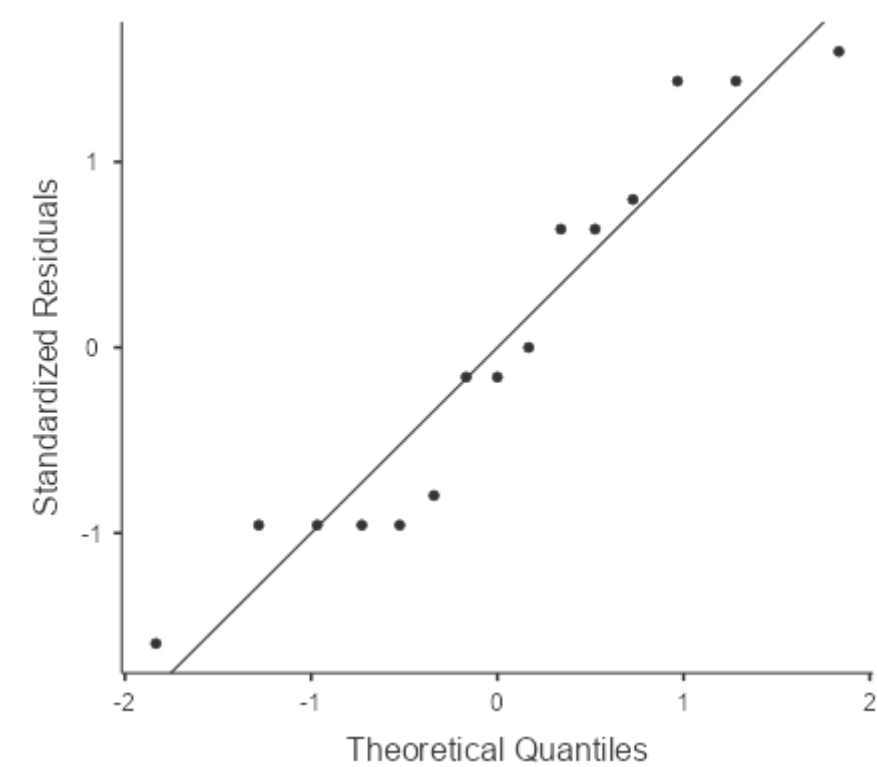
Note. Additional results provided by *moretests*

Normality tests

|                    | statistic | p     |
|--------------------|-----------|-------|
| Shapiro-Wilk       | 0.917     | 0.171 |
| Kolmogorov-Smirnov | 0.179     | 0.720 |
| Anderson-Darling   | 0.517     | 0.159 |

Note. Additional results provided by *moretests*

Q-Q Plot



Post Hoc Tests

Post Hoc Comparisons - Dose

| Comparison |         | Mean Difference | SE    | df   | t     | P <sub>tukey</sub> | Cohen's d |
|------------|---------|-----------------|-------|------|-------|--------------------|-----------|
| Dose       | Dose    |                 |       |      |       |                    |           |
| Control    | 15 mins | -1.00           | 0.887 | 12.0 | -1.13 | 0.516              | -0.713    |
|            | 30 mins | -2.80           | 0.887 | 12.0 | -3.16 | 0.021              | -1.997    |
| 15 mins    | 30 mins | -1.80           | 0.887 | 12.0 | -2.03 | 0.147              | -1.284    |

Note. Comparisons are based on estimated marginal means

# Robust ANOVA

Robust ANOVA

|      | F    | df1  | df2  | p     | ES    | Bootstrap CI |       |
|------|------|------|------|-------|-------|--------------|-------|
|      |      |      |      |       |       | Lower        | Upper |
| Dose | 3.00 | 2.00 | 4.00 | 0.160 | 0.789 | 0.434        | 1.49  |

Note. Method of trimmed means (level 0.2).  
Note. For effect size CI computation (samples 599)

## Post Hoc Tests

Post Hoc Tests - Dose

|         |         |       |       | 95% Confidence interval |                   |
|---------|---------|-------|-------|-------------------------|-------------------|
|         |         |       |       | Lower                   | Upper             |
| Control | 15 mins | -1.00 | 0.435 | -5.32 <sup>a</sup>      | 3.32 <sup>a</sup> |
|         | 30 mins | -3.00 | 0.181 | -7.32 <sup>a</sup>      | 1.32 <sup>a</sup> |
| 15 mins | 30 mins | -2.00 | 0.317 | -6.32 <sup>a</sup>      | 2.32 <sup>a</sup> |

<sup>a</sup> CI are adjusted to control FWE, but not p-values.

# ANOVA

ANOVA - ...

|           | Sum of Squares | df | Mean Square | F | p |
|-----------|----------------|----|-------------|---|---|
| ...       | .              | .  | .           | . | . |
| Residuals | .              | .  | .           | . | . |

[3]

## One-Way ANOVA

One-Way ANOVA (Welch's)

|           | F    | df1 | df2  | p     |
|-----------|------|-----|------|-------|
| Happiness | 4.32 | 2   | 7.94 | 0.054 |

## References

[1] The jamovi project (2024). *jamovi*. (Version 2.6) [Computer Software]. Retrieved from <https://www.jamovi.org>.

[2] R Core Team (2024). *R: A Language and environment for statistical computing*. (Version 4.4) [Computer software]. Retrieved from <https://cran.r-project.org>. (R packages retrieved from CRAN snapshot 2024-08-07).

**[3]** Fox, J., & Weisberg, S. (2023). *car: Companion to Applied Regression*. [R package]. Retrieved from <https://cran.r-project.org/package=car>.

**[4]** Lenth, R. (2023). *emmeans: Estimated Marginal Means, aka Least-Squares Means*. [R package]. Retrieved from <https://cran.r-project.org/package=emmeans>.